



Quantum computing

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Content of course

- Theoretical basis for quantum computing.
- Practical basis for quantum computing.
- Near-term potential of quantum computing.
- Other applications of quantum information.

Quantum computing properties

Re-cap: measurement in the computational basis

If we have a state $|\psi\rangle$ which is either $|0\rangle$ or $|1\rangle$, then we can perfectly distinguish which of these it is by measurement in the computational basis:

$$|1\rangle = 0|0\rangle + 1|1\rangle$$

so we measure 1 with probability $|1|^2 = 1$ (and likewise for 0).

Essentially, this is just classical (binary) information.

Quantum computing properties

Distinguishing any pair of orthogonal states

If we now have a state $|\psi\rangle$ which is either $|\psi_0\rangle = \alpha|0\rangle + \beta|1\rangle$ or $|\psi_1\rangle = \gamma|0\rangle + \delta|1\rangle$, such that $|\psi_0\rangle$ and $|\psi_1\rangle$ are orthogonal, then we can still perfectly distinguish which of these it is by first noting that the following matrix is unitary (i.e., because the columns form an orthonormal basis):

$$U = \begin{bmatrix} \alpha & \gamma \\ \beta & \delta \end{bmatrix}$$

and observing that the transform $U^\dagger$ sends $|\psi_0\rangle \rightarrow |0\rangle$ and $|\psi_1\rangle \rightarrow |1\rangle$, which can thus be followed by a computational basis measurement to discover the state.

In fact, physicists and mathematicians frequently speak not of doing a transformation such that the states are aligned with the computational basis, but rather performing the measurement in the basis $(|\psi_0\rangle, |\psi_1\rangle)$, which amounts to the same thing.

Quantum computing properties

Measurement in the $(|\psi_0\rangle, |\psi_1\rangle)$ basis

Measurement in the $(|\psi_0\rangle, |\psi_1\rangle)$ basis means measurement with the following operators:

$$M_{|\psi_0\rangle} = |\psi_0\rangle \langle\psi_0| \quad M_{|\psi_1\rangle} = |\psi_1\rangle \langle\psi_1|$$

Using the fact that projectors are self-adjoint, and for any $|\psi\rangle$:

$$(|\psi\rangle \langle\psi|)^2 = |\psi\rangle \langle\psi| |\psi\rangle \langle\psi| = |\psi\rangle \langle\psi| |\psi\rangle \langle\psi| = |\psi\rangle \times 1 \times \langle\psi| = |\psi\rangle \langle\psi|$$

we can see that measurement in the $(|\psi_0\rangle, |\psi_1\rangle)$ basis fulfils the completeness equation:

$$\begin{aligned} \sum_m M_m^\dagger M_m &= M_{|\psi_0\rangle}^\dagger M_{|\psi_0\rangle} + M_{|\psi_1\rangle}^\dagger M_{|\psi_1\rangle} \\ &= (|\psi_0\rangle \langle\psi_0|)^2 + (|\psi_1\rangle \langle\psi_1|)^2 \\ &= |\psi_0\rangle \langle\psi_0| + |\psi_1\rangle \langle\psi_1| \\ &= U |0\rangle \langle 0| U^\dagger + U |1\rangle \langle 1| U^\dagger \\ &= U (|0\rangle \langle 0| + |1\rangle \langle 1|) U^\dagger \\ &= U I U^\dagger \\ &= I \end{aligned}$$

(with U as defined on the previous slide).

Quantum computing properties

Measurement in the $(|\psi_0\rangle, |\psi_1\rangle)$ basis (continued)

In the previous slide we showed that single-qubit measurement in any orthogonal basis satisfies the completeness equation. We will now verify that measurement in the $(|\psi_0\rangle, |\psi_1\rangle)$ basis perfectly distinguishes $(|\psi_0\rangle$ and $|\psi_1\rangle)$. Let $|\psi\rangle$ be some state that is either $(|\psi_0\rangle$ or $|\psi_1\rangle)$, we have:

$$p(M_{|\psi_0\rangle} | |\psi\rangle = |\psi_0\rangle) = \langle\psi_0| (|\psi_0\rangle \langle\psi_0|)^\dagger |\psi_0\rangle \langle\psi_0| |\psi_0\rangle = (\langle\psi_0|\psi_0\rangle)^3 = 1$$

$$p(M_{|\psi_1\rangle} | |\psi\rangle = |\psi_1\rangle) = \langle\psi_1| (|\psi_1\rangle \langle\psi_1|)^\dagger |\psi_1\rangle \langle\psi_1| |\psi_1\rangle = (\langle\psi_1|\psi_1\rangle)^3 = 1$$

and also (to confirm):

$$\begin{aligned} p(M_{|\psi_0\rangle} | |\psi\rangle = |\psi_1\rangle) &= \langle\psi_1| (|\psi_0\rangle \langle\psi_0|)^\dagger |\psi_0\rangle \langle\psi_0| |\psi_1\rangle \\ &= \langle\psi_1|\psi_0\rangle \langle\psi_0|\psi_0\rangle \langle\psi_0|\psi_1\rangle \\ &= 0 \end{aligned}$$

$$\begin{aligned} p(M_{|\psi_1\rangle} | |\psi\rangle = |\psi_0\rangle) &= \langle\psi_0| (|\psi_1\rangle \langle\psi_1|)^\dagger |\psi_1\rangle \langle\psi_1| |\psi_0\rangle \\ &= \langle\psi_0|\psi_1\rangle \langle\psi_1|\psi_1\rangle \langle\psi_1|\psi_0\rangle \\ &= 0 \end{aligned}$$

i.e., because $|\psi_0\rangle$ and $|\psi_1\rangle$ are orthogonal, so $\langle\psi_0|\psi_1\rangle = \langle\psi_1|\psi_0\rangle = 0$.

Quantum computing properties

It is not possible to perfectly distinguish non-orthogonal quantum states

If we now have a state $|\psi\rangle$ which is either $|\psi_a\rangle$ or $|\psi_b\rangle$ which are **not orthogonal**, then there is no measurement that can perfectly tell us which of these states $|\psi\rangle$ is.

... but we can perform a measurement that tells us something about the likelihood of whether $|\psi\rangle = |\psi_a\rangle$ or $|\psi\rangle = |\psi_b\rangle$.

Intuitively:

- If we just guess, we will be correct with probability equal to one half, so we expect to be able to do better than this.
- The “closer together” $|\psi_a\rangle$ and $|\psi_b\rangle$ are, the harder they will be to distinguish (i.e., the lower the probability of correctly inferring $|\psi\rangle$)

Quantum computing properties

The Helstrom-Holevo Bound

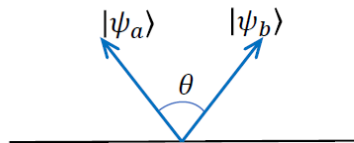
The intuition in the previous slide turns out to be correct, and is captured by the Helstrom-Holevo bound:

If $|\psi\rangle$ is either $|\psi_a\rangle$ or $|\psi_b\rangle$, where $|\langle\psi_a|\psi_b\rangle| = \cos\theta$, then the probability of correctly inferring the state $|\psi\rangle$ is less than or equal to $\frac{1}{2}(1 + \sin\theta)$.

Furthermore, the bound is tight, it can always be achieved by choosing the measurement basis as the eigenvectors of:

$$|\psi_a\rangle\langle\psi_a| - |\psi_b\rangle\langle\psi_b|$$

This can be visualised:



So we can see that, if $|\psi_a\rangle$ and $|\psi_b\rangle$ are orthogonal (i.e., $\theta = \pi/2$), then $\sin\theta = 1$ and so they can be perfectly distinguished if the correct measurement basis is selected (i.e., because $\frac{1}{2}(1 + \sin\theta) = 1$). Conversely, if $|\psi_a\rangle$ and $|\psi_b\rangle$ are nearly aligned, so $\theta \approx 0$, then $\sin\theta \approx 0$ and $\frac{1}{2}(1 + \sin\theta) \approx \frac{1}{2}$: so we cannot do better than guessing.

Quantum computing properties

Example: distinguishing $|0\rangle$ and $|+\rangle$

First, we note that $|0\rangle$ and $|+\rangle$ are not orthogonal:

$$\begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} = \frac{1}{\sqrt{2}} \neq 0$$

therefore we cannot perfectly infer the state of $|\psi\rangle$ if we know it is either $|0\rangle$ or $|+\rangle$. So instead, we must decide a basis to measure in:

$$|0\rangle\langle 0| - |+\rangle\langle +| = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & -\frac{1}{2} \end{bmatrix}$$

which has eigenvectors $[0.38 \ 0.92]^T$ and $[-0.92 \ 0.38]^T$. Finally, we can calculate the probability of correctly inferring the state:

$$\frac{1}{2}(1 + \sin(\arccos(1/\sqrt{2}))) = 0.85$$

Quantum computing properties

Distinguishing $|0\rangle$ and $|+\rangle$ by measuring in the computational basis

Again we have that $|\psi\rangle$ is either $|0\rangle$ or $|+\rangle$, each with 50% probability. We can tabulate the quantum states and measurement outcomes when measuring in the computational basis:

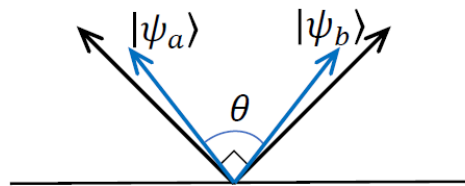
	0	1
$ 0\rangle$	$\frac{1}{2}$	0
$ +\rangle$	$\frac{1}{4}$	$\frac{1}{4}$

- $\frac{1}{4}$ of the time we will measure 1, which means $|\psi\rangle = |+\rangle$.
- $\frac{3}{4}$ of the time we will measure 0, which we should *guess* means $|\psi\rangle = |0\rangle$, but of these $\frac{1}{3}$ will be wrong, and actually $|\psi\rangle = |+\rangle$.
- So we have success probability $1 - \frac{3}{4} \times \frac{1}{3} = \frac{3}{4}$, which is less than the theoretically achievable 0.85, but if we measure 1 then we know $|\psi\rangle = |+\rangle$ with certainty.

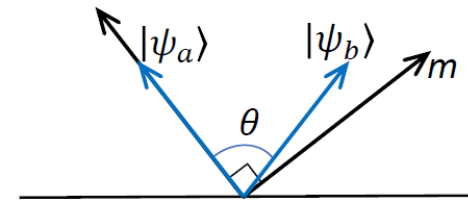
Quantum computing properties

Depicting different state discrimination strategies

The longer black arrows show the orthogonal measurement basis; the shorter blue arrows show the non-orthogonal states that we wish to distinguish.



The optimal strategy is to choose the measurement basis spread equally each side of the states we want to distinguish.



But if one of the measurement basis vectors aligns with one of the states being distinguished, sometimes we get a measurement that we are 100% sure about.

In the right-hand plot, if we measure the outcome corresponding to the right-hand black arrow (marked m), then we know the state is $|\psi_b\rangle$, because $|\psi_a\rangle$ is orthogonal to this measurement basis vector.

Quantum computing properties

The no-signalling principle: why it matters

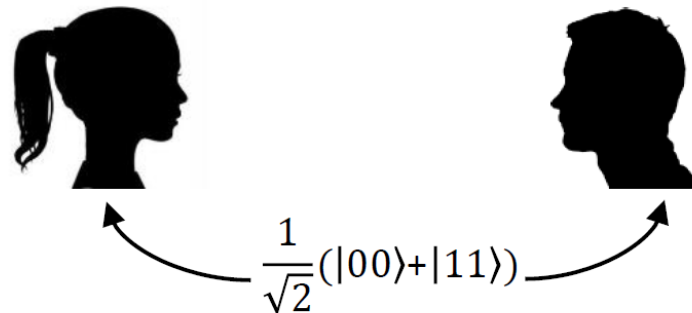
We have met the concept of entanglement, for example the Bell state $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ is an entangled state (sometimes termed a *Bell pair* or *EPR pair*). This has the property that if we **measure the first qubit** (in the computational basis) then we either get measurement outcome **0**, in which case the state collapses to $|00\rangle$; or measurement outcome **1**, in which case the state collapses to $|11\rangle$.

Notably, even though the **second qubit** hasn't been touched, its state has still be collapsed by virtue of the measurement on the first qubit. There is no physical requirement that the two entangled qubits are in close proximity to each other (*local* in physics parlance), but this collapse happens to both qubits instantaneously upon the measurement of the first qubit. This then implies some non-local action on the second qubit: its state has instantly changed as a result of a distant action. This is what Einstein referred to as “spooky action at a distance”, but what is really important is whether this can be used to transfer information faster than the speed-of-light, which would violate the theory of relativity.

In fact, collapsing entanglement in this way *cannot* be used to transfer information, as proven by the no-signalling principle.

Quantum computing properties

The no-signalling principle: set-up



- Alice and Bob are at different ends of the universe, but each have one half of a Bell pair: $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$.
- Alice can measure her qubit whenever she wants, and this will collapse Bob's to the same state.
- We are interested in whether Bob can infer whether or not Alice has measured her qubit.
- But all that Bob can do to infer whether Alice has measured her qubit is to measure his own qubit – therefore, the question reduces to whether the measurement probabilities that Bob sees are altered by virtue of Alice having performed her measurement.

Quantum computing properties

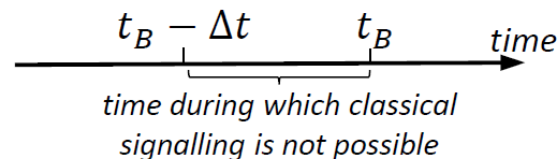
The no-signalling principle: set-up (continued)

If Bob can infer from measurement of his qubit whether or not Alice has measured hers, then this *does* enable super-luminal (faster than the speed-of-light) information transfer. Consider the following set-up.

- Alice and Bob are spatially separated by a distance that takes light Δt seconds to traverse.
- Bob is interested in whether some event that Alice witnesses has occurred before time t_B .
- When Alice witnesses the event she will signal to notify Bob.

So we have two alternatives:

1. If Alice uses classical signalling, then if the event occurs less than Δt seconds before t_B , then there is no way she can send a signal to Bob that he will receive before t_B .
2. However, if Alice can send a signal solely by measuring her qubit, then she can signal instantly, and hence notify Bob of the event any time up to t_B .



Quantum computing properties

The no-signalling principle: proof

If Alice hasn't measured her qubit, then the state is $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$, and so Bob has a $\left(\frac{1}{\sqrt{2}}\right)^2 = \frac{1}{2}$ probability of measuring each of 0 and 1.

If Alice has measured her qubit, then Bob's qubit has collapsed – it is either in state 0, or state 1 (each with probability $1/2$). However, in the absence of signalling, Bob has no knowledge of which of these measurement outcomes Alice observed, and so all he knows is that he will measure each of $|0\rangle$ and $|1\rangle$ with probability $1/2$. So the no-signalling principle is proved.

- Crucially, in the absence of signalling, Bob's measurement statistics when measuring the uncollapsed quantum state are identical to his lack of knowledge (expressed probabilistically) when measuring the state already collapsed by Alice.
- The no-signalling principle also holds for any type of entanglement, and also any scheme Alice and Bob may come up with involving transformations of their qubits, and measurements in arbitrary bases.

Quantum computing properties

The no-cloning principle: why it matters

- A plethora of physics reasons.
- That we cannot clone makes quantum error-correction harder.
- The possibility of cloning would enable the violation of the no-signalling principle (see exercise sheet).
- Cloning would enable an infinite amount of classical information to be compressed into a single qubit and then recovered afterwards:
 1. Map a classical bit-string to a unique qubit state.
 2. Communicate the single qubit.
 3. Receive the qubit, make an arbitrary number of copies by cloning, and perform quantum state tomography to recover the original classical information.

Quantum computing properties

The no-cloning principle: set-up

We have a quantum state $|\psi\rangle$ and a register initially set to $|0\rangle$, and we wish to find a cloning unitary, U such that:

$$U(|\psi\rangle|0\rangle) = |\psi\rangle|\psi\rangle$$

We will now prove that no such U exists.

Quantum computing properties

The no-cloning principle: proof

Consider that U must clone all quantum states, so as well as

$$U(|\psi\rangle|0\rangle) = |\psi\rangle|\psi\rangle$$

from the previous slide, we have that

$$U(|\phi\rangle|0\rangle) = |\phi\rangle|\phi\rangle$$

Taking the inner products of the left- and right-hand sides of the above equations, we have that:

$$\begin{aligned}\langle\psi|\langle 0|U^\dagger U|\phi\rangle|0\rangle &= (\langle\psi|\langle\psi\rangle)(|\phi\rangle|\phi\rangle) \\ \implies \langle\psi|\phi\rangle\langle 0|0\rangle &= (\langle\psi|\phi\rangle)^2 \\ \implies \langle\psi|\phi\rangle &= (\langle\psi|\phi\rangle)^2\end{aligned}$$

which is only true if $\psi = \phi$ or ψ and ϕ are orthogonal (so their inner-product is 0). So we have proven that there exists no unitary U that can clone arbitrary quantum states.

Quantum computing properties

The no-deleting principle

Time-reversal of the no-cloning principle yields the no-deleting principle: there does not exist a unitary $\tilde{U}$ that can delete one of two copies of a quantum state, that is:

$$\tilde{U}(|\psi\rangle |\psi\rangle) = |\psi\rangle |0\rangle$$

It is less obvious why this is useful, but the no-deleting principle does arise in quantum information, and so it is worth being aware of.

More generally, quantum computing is reversible (except for measurement), and therefore the (im)possibility of some computation implies the (im)possibility of its reverse.